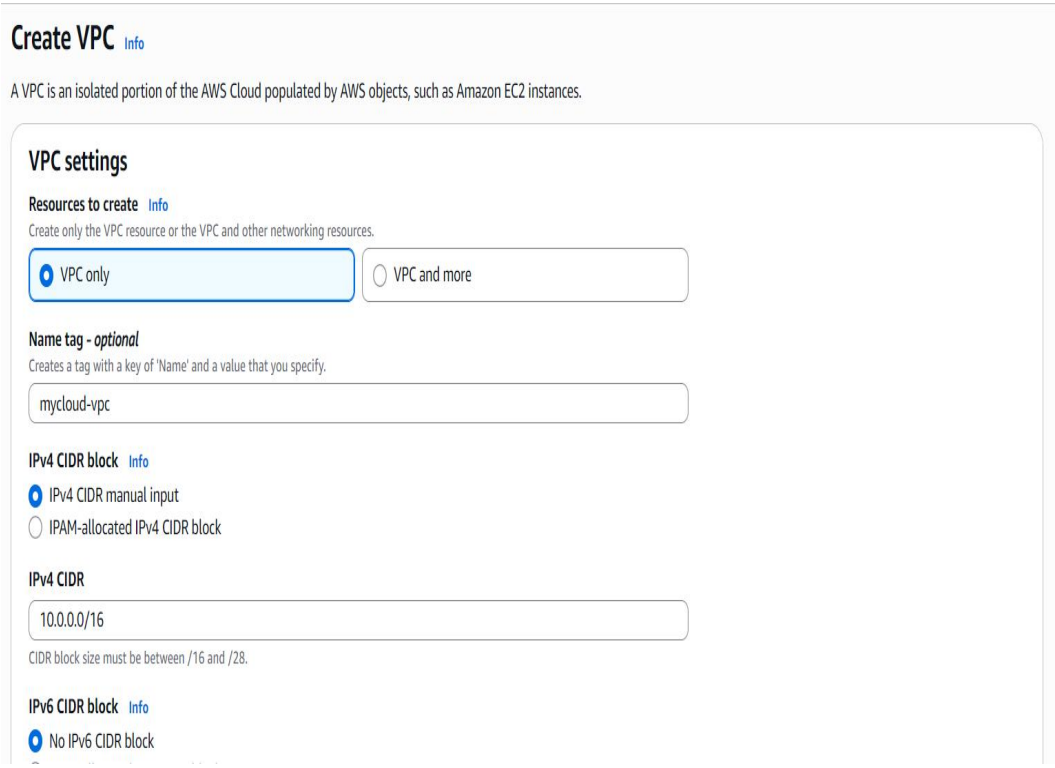
Build a Virtual Private Cloud (VPC)

Abhijeet Pandit



Introduction Today's Project!

What is Amazon VPC?

Amazon VPC lets you create a private network in AWS where you control IP ranges, subnets, routing , and security. It's useful for safely running resources like EC2, with full control over internet access and traffic within your cloud setup.

How I used Amazon VPC in this project

I used Amazon VPC in today's project to set up a secure network for my resources. I created subnets in the Mumbai region. Enabled auto-assign public IPs, and attached an internet gateway so instances could access the internet. This gave me full control.

One Thing I didn't expect in this project was…

Another AWS Service done ! One thing I didn't expect in this project was that just naming a subnet "Public" doesn't actually make it public. I had to manually add a route to the internet gateway something that's easy to miss but critical for internet.

This project took me…

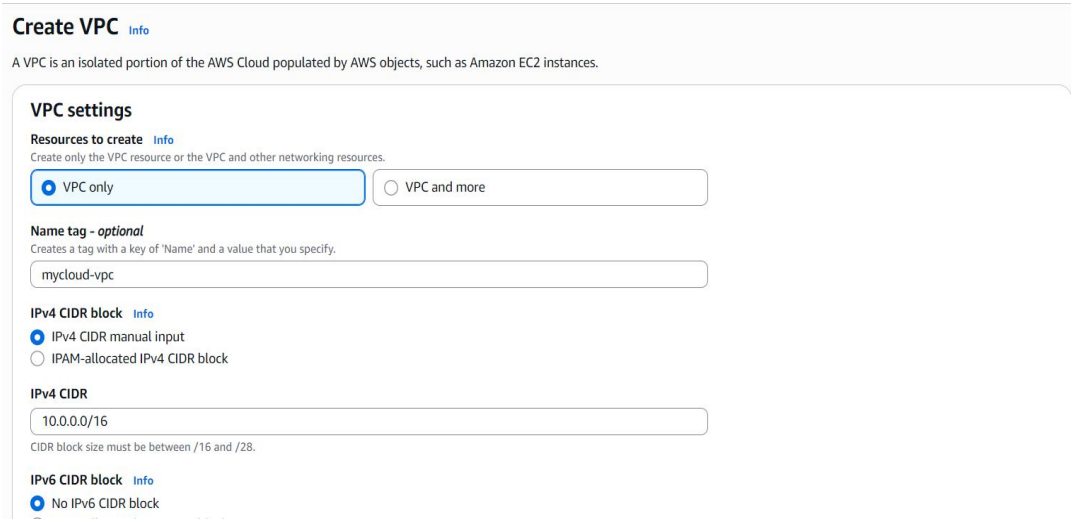
This Project took me about 1 to 2 hours. Most of the time went into understanding how VPC components like subnets, route tables, and internet gateways work together. Once I got the hang of it, setting things up was pretty smooth!

Virtual Private Clouds (VPC)

A VPC (Virtual Private Cloud) is a private network in the cloud. VPC is like your own private space in the cloud. You choose who and what can connect to it, kind of like setting up Wi-Fi at home with guest access, security rules, and rooms (subnets)

There was already a default VPC in my account ever since my AWS account was created. This is because AWS has set up a default VPC to allow me to deploy resources like EC2 instance/RDS databases right away (without creating my own VPC from scratch).

To set up my VPC, I had to define an IPv4 CIDR block, which is means a range of IP addresses that my VPC can allocate to the resources deplyed into my VPC

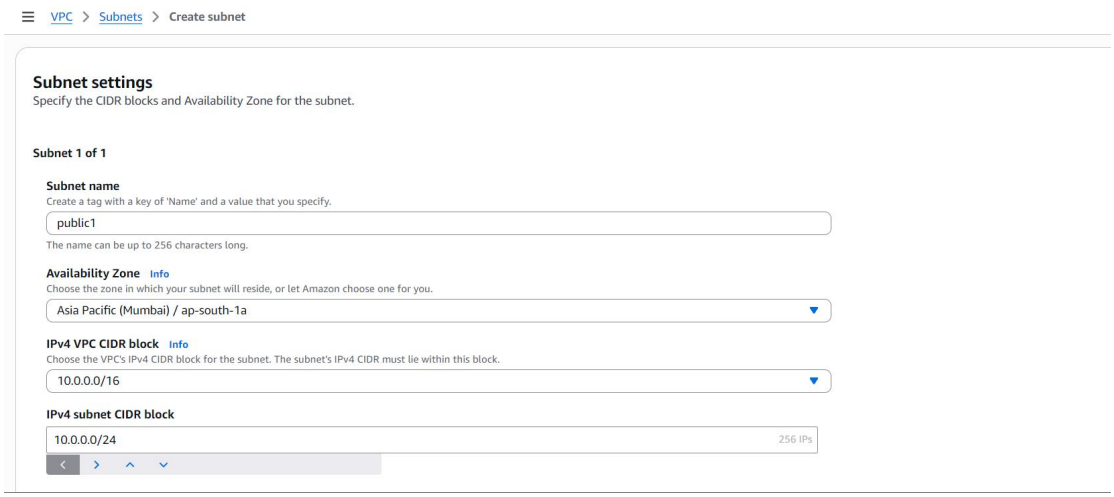


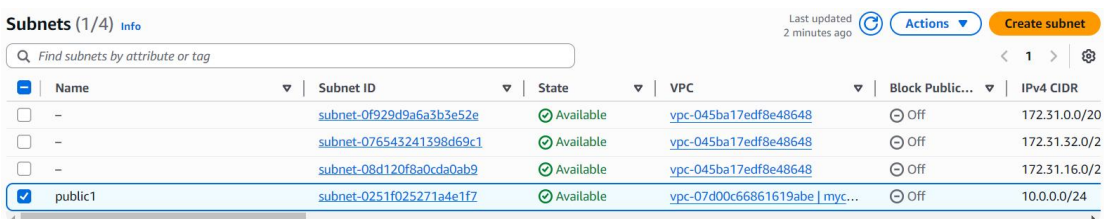
Subnets

Subnets are the smaller sections inside your VPC, kind of like neighborhoods in a city. There are already subnets existing in my account, one for every Availability Zone in my region. Since I'm using the Mumbai region, I have three subnets by default

Once I created my subnet, I enabled auto-assign publice IPv4 addresses. This setting makes sure any instance I launch in that subnet gets a public IP, so that it can connect to the internet right away without needing extra setup.

The difference between publice and private subnets are about internet access. For a subnet to be considered publice, it has to be connected to something called an internet gateway. Without the connection, the subnet can't reach the internet.





Internet gateways

Internet gateways are components that allow communication between instances in your VPC and the internet. They let publice subnets send and receive data from the internet, acting like a bridge between ypur cloud network and the outside world.

Attaching an internet gateway to a VPC means resources in public subnets can access the internet. If I missed this step, even if instances have publice IPs, they won't be able to connect to or be reached from the internet.

Internet gateways (2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State
–	igw-03d318ce441d465ca	Attached
mygateway1	igw-0860bb3c138bfe47d	Attached